Proof by Mathematical Induction

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The three steps of induction**

State the three steps required for a valid proof by mathematical induction.

(1) Base case: verify the statement holds for the smallest value of n (usually $n = 1$). (2) Inductive hypothesis: assume the statement holds for $n = k$. (3) Inductive step: use the assumption to prove the statement holds for $n = k + 1$. Concluding, by the principle of mathematical induction, the statement holds for all n in the given range.

2 **[Foundational]** • **Target: Proving a summation formula**

Prove by induction that $1 + 2 + 3 + \cdots + n = n(n + 1)/2$ for all $n \geq 1$.

Base case ($n = 1$): LHS $= 1$. RHS $= 1(2)/2 = 1$. True. Inductive hypothesis: assume $1 + 2 + \cdots + k = k(k + 1)/2$. For $n = k + 1$: $1 + 2 + \cdots + k + (k + 1) = k(k + 1)/2 + (k + 1)$. Factoring out $(k + 1)$: this is $(k + 1)(k/2 + 1) = (k + 1)(k + 2)/2$, which matches the formula with $n = k + 1$. Since it holds for $n = 1$ and $n = k \Rightarrow n = k + 1$, by induction it holds for all $n \geq 1$.

3 **[Foundational]** • **Target: Proving a summation formula**

Prove by induction that $1^2 + 2^2 + 3^2 + \cdots + n^2 = n(n + 1)(2n + 1)/6$ for all $n \geq 1$.

Base case ($n = 1$): LHS $= 1$. RHS $= 1(2)(3)/6 = 1$. True. Assume true for $n = k$: $1^2 + 2^2 + \cdots + k^2 = k(k + 1)(2k + 1)/6$. For $n = k + 1$: $1^2 + \cdots + k^2 + (k + 1)^2 = k(k + 1)(2k + 1)/6 + (k + 1)^2$. Factoring out $(k + 1)$: this is $(k + 1)[k(2k + 1)/6 + (k + 1)]$. Combining the bracketed terms over a denominator of 6: this is $(k + 1)(2k^2 + 7k + 6)/6$. Since $2k^2 + 7k + 6 = (2k + 3)(k + 2)$, this equals $(k + 1)(k + 2)(2k + 3)/6$, matching the formula at $n = k + 1$. By induction, true for all $n \geq 1$.

4 **[Foundational]** • **Target: Proving divisibility results**

Prove by induction that $7^n - 1$ is divisible by 6 for all $n \geq 1$.

Base case ($n = 1$): $7^1 - 1 = 6$, divisible by 6. Assume $7^k - 1 = 6m$ for some integer m, so $7^k = 6m + 1$. For $n = k + 1$: $7^{k+1} - 1 = 7 \cdot 7^k - 1 = 7(6m + 1) - 1 = 42m + 6 = 6(7m + 1)$, which is divisible by 6. By induction, $7^n - 1$ is divisible by 6 for all $n \geq 1$.

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5 [Foundational] • Target: Induction with inequalities

Prove by induction that $2^n > n^2$ for all integers $n \geq 5$.

Base case ($n = 5$): $2^5 = 32$, $5^2 = 25$, and $32 > 25$. True. Assume $2^k > k^2$ for some $k \geq 5$. For $n = k + 1$: $2^{k+1} = 2 \cdot 2^k > 2k^2$. We need $2k^2 \geq (k+1)^2 = k^2 + 2k + 1$, i.e. $k^2 - 2k - 1 \geq 0$, which holds for $k \geq 3$ (true since $k \geq 5$). So $2^{k+1} > 2k^2 \geq (k+1)^2$. By induction, $2^n > n^2$ for all $n \geq 5$.

6 [Foundational] • Target: Induction in calculus: derivatives of a product

Prove by induction that $\frac{d^n}{dx^n}(e^{ax}) = a^n e^{ax}$ for all positive integers n.

Base case ($n = 1$): differentiating e^{ax} once gives $ae^{ax} = a^1 e^{ax}$, matching the formula. Inductive hypothesis: assume the k-th derivative of e^{ax} is $a^k e^{ax}$. Differentiating this once more: $\frac{d}{dx}(a^k e^{ax}) = a^k \cdot ae^{ax} = a^{k+1} e^{ax}$ — this is exactly the $(k+1)$-th derivative, matching the formula at $n = k + 1$. By induction, true for all positive integers n.

7 [Foundational] • Target: Induction with a recursively defined sequence

A sequence is defined by $u_1 = 3$ and $u_{n+1} = 2u_n - 1$. Prove by induction that $u_n = 2^n + 1$ for all $n \geq 1$.

Base case ($n = 1$): $u_1 = 3$ (given). Formula: $2^1 + 1 = 3$. True. Assume $u_k = 2^k + 1$. Then $u_{k+1} = 2u_k - 1 = 2(2^k + 1) - 1 = 2^{k+1} + 2 - 1 = 2^{k+1} + 1$, matching the formula at $n = k + 1$. By induction, $u_n = 2^n + 1$ for all $n \geq 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Proving a geometric-series-style summation

Prove by induction that the sum $3^0 + 3^1 + \cdots + 3^{n-1}$ equals $(3^n - 1)/2$ for all $n \geq 1$.

Base case ($n = 1$): LHS $= 3^0 = 1$. RHS $= (3^1 - 1)/2 = 2/2 = 1$. True. Assume the sum $3^0 + \cdots + 3^{k-1} = (3^k - 1)/2$. For $n = k + 1$: adding the next term 3^k gives $(3^k - 1)/2 + 3^k$. Combining over a denominator of 2: this is $(3^k - 1 + 2 \cdot 3^k)/2 = (3 \cdot 3^k - 1)/2 = (3^{k+1} - 1)/2$, matching the formula at $n = k + 1$. By induction, true for all $n \geq 1$.

9 [Intermediate] • Target: Proving a geometric-series-style summation

Prove by induction that the sum $1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n!$ equals $(n+1)! - 1$ for all $n \geq 1$.

Base case ($n = 1$): LHS $= 1 \cdot 1! = 1$. RHS $= 2! - 1 = 2 - 1 = 1$. True. Assume $1 \cdot 1! + \cdots + k \cdot k! = (k+1)! - 1$. For $n = k + 1$: adding the next term $(k+1)(k+1)!$ gives $(k+1)! - 1 + (k+1)(k+1)! = (k+1)![1 + (k+1)] - 1 = (k+1)!(k+2) - 1 = (k+2)! - 1$, matching the formula at $n = k + 1$. By induction, true for all $n \geq 1$.

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10 **[Intermediate]** • **Target: Proving an inequality involving factorials**

Prove by induction that $n! > 2^n$ for all integers $n \geq 4$.

Base case ($n = 4$): $4! = 24$, $2^4 = 16$, and $24 > 16$. True. Assume $k! > 2^k$ for some $k \geq 4$. For $n = k + 1$: $(k+1)! = (k+1) \cdot k! > (k+1) \cdot 2^k$. Since $k \geq 4$, $k + 1 \geq 5 > 2$, so $(k+1) \cdot 2^k > 2 \cdot 2^k = 2^{k+1}$. Combining, $(k+1)! > 2^{k+1}$, matching the inequality at $n = k + 1$. By induction, $n! > 2^n$ for all $n \geq 4$.

11 **[Intermediate]** • **Target: Proving a divisibility result with two variables**

Prove by induction that $n^3 + 2n$ is divisible by 3 for all positive integers $n \geq 1$.

Base case ($n = 1$): $1^3 + 2(1) = 3$, divisible by 3. Assume $k^3 + 2k = 3m$ for some integer m. For $n = k + 1$: $(k+1)^3 + 2(k+1) = k^3 + 3k^2 + 3k + 1 + 2k + 2$. Grouping terms: this is $(k^3 + 2k) + (3k^2 + 3k + 3) = 3m + 3(k^2 + k + 1)$. Factoring out 3: $3m + 3(k^2 + k + 1) = 3(m + k^2 + k + 1)$, which is divisible by 3. By induction, $n^3 + 2n$ is divisible by 3 for all $n \geq 1$.

12 **[Intermediate]** • **Target: Proving a divisibility result with two variables**

Prove by induction that $9^n - 1$ is divisible by 8 for all positive integers $n \geq 1$, and explain briefly why a similar argument would show $a^n - 1$ is divisible by $a - 1$ for any integer $a > 1$.

Base case ($n = 1$): $9^1 - 1 = 8$, divisible by 8. Assume $9^k - 1 = 8m$ for some integer m, so $9^k = 8m + 1$. For $n = k + 1$: $9^{k+1} - 1 = 9 \cdot 9^k - 1 = 9(8m + 1) - 1 = 72m + 9 - 1 = 72m + 8 = 8(9m + 1)$, divisible by 8. By induction, true for all $n \geq 1$. The same inductive pattern works for any base a: $a^{k+1} - 1 = a \cdot a^k - 1 = a[(a-1)m' + 1] - 1 = (a-1)(am') + (a-1) = (a-1)(am' + 1)$, always a multiple of $a - 1$.

13 **[Intermediate]** • **Target: Induction with a recurrence relation for a sequence of sums**

A sequence satisfies $S_1 = 2$ and $S_{n+1} = S_n + 6n + 2$ for $n \geq 1$. Prove by induction that $S_n = 3n^2 - n$ for all $n \geq 1$.

Base case ($n = 1$): $S_1 = 2$ (given). Formula: $3(1)^2 - 1 = 2$. True. Assume $S_k = 3k^2 - k$. Then $S_{k+1} = S_k + 6k + 2 = 3k^2 - k + 6k + 2 = 3k^2 + 5k + 2$. The formula at $n = k + 1$ gives $3(k+1)^2 - (k+1) = 3k^2 + 6k + 3 - k - 1 = 3k^2 + 5k + 2$, which matches exactly. Since it holds for $n = 1$ and $n = k \Rightarrow n = k + 1$, by induction $S_n = 3n^2 - n$ for all $n \geq 1$.

14 **[Intermediate]** • **Target: Induction with a strengthened hypothesis**

Prove by induction that $11^n - 4^n$ is divisible by 7 for all positive integers $n \geq 1$.

Base case ($n = 1$): $11^1 - 4^1 = 7$, divisible by 7. Assume $11^k - 4^k = 7m$ for some integer m, so $11^k = 7m + 4^k$. For $n = k + 1$: $11^{k+1} - 4^{k+1} = 11 \cdot 11^k - 4 \cdot 4^k$. Substituting $11^k = 7m + 4^k$: this is $11(7m + 4^k) - 4 \cdot 4^k = 77m + 11 \cdot 4^k - 4 \cdot 4^k$. Combining the 4^k terms: $77m + 7 \cdot 4^k$. Factoring out 7: $7(11m + 4^k)$, which is divisible by 7. By induction, $11^n - 4^n$ is divisible by 7 for all $n \geq 1$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Induction to prove a formula for the sum of cubes** • **Cross-Lesson: 6.01 Matrices (AI HL)**

Prove by induction that $1^3 + 2^3 + 3^3 + \cdots + n^3 = [n(n+1)/2]^2$ for all $n \geq 1$.

Base case ($n = 1$): LHS $= 1^3 = 1$. RHS $= [1(2)/2]^2 = 1^2 = 1$. True. Assume $1^3 + \cdots + k^3 = [k(k+1)/2]^2$. For $n = k+1$: adding $(k+1)^3$ gives $[k(k+1)/2]^2 + (k+1)^3$. Factoring out $(k+1)^2$: this is $(k+1)^2[k^2/4 + (k+1)]$. Combining over a denominator of 4: this is $(k+1)^2(k^2+4k+4)/4$. Since $k^2+4k+4 = (k+2)^2$, this is $(k+1)^2(k+2)^2/4 = [(k+1)(k+2)/2]^2$, matching the formula at $n = k+1$. By induction, true for all $n \geq 1$.

16 **[Challenging]** • **Target: Induction to prove a formula for the sum of cubes** • **Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)**

Prove by induction that $5^n + 2 \cdot 3^{n-1} + 1$ is divisible by 8 for all positive integers $n \geq 1$.

Base case ($n = 1$): $5^1 + 2(3^0) + 1 = 5 + 2 + 1 = 8$, divisible by 8. Assume $5^k + 2 \cdot 3^{k-1} + 1 = 8m$ for some integer m. Consider the difference between consecutive terms: $(5^{k+1} + 2 \cdot 3^k + 1) - (5^k + 2 \cdot 3^{k-1} + 1) = 5^{k+1} - 5^k + 2 \cdot 3^k - 2 \cdot 3^{k-1}$. This simplifies to $4 \cdot 5^k + 4 \cdot 3^{k-1} = 4(5^k + 3^{k-1})$. Since 5^k and 3^{k-1} are both odd, their sum is even, so $4(5^k + 3^{k-1})$ is a multiple of 8. Therefore $5^{k+1} + 2 \cdot 3^k + 1 = 8m + 4(5^k + 3^{k-1})$, and since both terms are multiples of 8, so is their sum..

17 **[Challenging]** • **Target: Induction to prove a formula for a weighted geometric sum** • **Cross-Lesson: 6.03 Markov Chains (AI HL)**

Prove by induction that $1 \cdot 2^1 + 2 \cdot 2^2 + 3 \cdot 2^3 + \cdots + n \cdot 2^n = (n-1)2^{n+1} + 2$ for all $n \geq 1$.

Base case ($n = 1$): LHS $= 1 \cdot 2^1 = 2$. RHS $= (1-1)2^2 + 2 = 0 + 2 = 2$. True. Assume $1 \cdot 2^1 + \cdots + k \cdot 2^k = (k-1)2^{k+1} + 2$. For $n = k+1$: adding the next term $(k+1)2^{k+1}$ gives $(k-1)2^{k+1} + 2 + (k+1)2^{k+1}$. Combining the 2^{k+1} terms: this is $[(k-1) + (k+1)]2^{k+1} + 2 = 2k \cdot 2^{k+1} + 2 = k \cdot 2^{k+2} + 2$. The formula at $n = k+1$ requires $[(k+1) - 1]2^{k+2} + 2 = k \cdot 2^{k+2} + 2$, which matches exactly. By induction, the formula holds for all $n \geq 1$.

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18 **[Challenging]** • **Target: A closing check: identifying an invalid induction argument** • **Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra**

A student claims to prove that all horses are the same color using induction on the number of horses n in a group: the base case $n = 1$ is trivially true (one horse is trivially "the same color" as itself), and the student then argues that if any k horses are the same color, then any $k + 1$ horses must also be the same color. Explain precisely where this argument breaks down, and state why a correct induction proof must avoid this flaw.

The flaw is in the inductive step: the argument tries to overlap two groups of k horses (the first k and the last k out of $k + 1$ total) and conclude they share a common color through the overlap. This overlap argument only works when the overlap is non-empty, which requires $k \geq 2$. It fails silently at the transition from $n = 1$ to $n = 2$, since two groups of 1 horse each (drawn from a total of 2) have no horse in common, so there is no shared reference point linking their colors. This illustrates why a c.

19 **[Challenging]** • **Target: Proving a sum-of-products formula by induction** • **Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity**

Prove by induction that $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n(n+1) = n(n+1)(n+2)/3$ for all $n \geq 1$.

Base case $n = 1$: LHS $= 1 \cdot 2 = 2$, RHS $= 1(2)(3)/3 = 2$. Equal, so the base case holds. Inductive step: assume $1 \cdot 2 + \cdots + m(m+1) = m(m+1)(m+2)/3$ for some $m \geq 1$. Adding $(m+1)(m+2)$ to both sides: the new sum is $m(m+1)(m+2)/3 + (m+1)(m+2)$. Factoring out $(m+1)(m+2)$: this is $(m+1)(m+2)[m/3 + 1] = (m+1)(m+2)(m+3)/3$ — exactly the claimed formula with $n = m + 1$. By induction, the formula holds for all $n \geq 1$.

20 **[Challenging]** • **Target: Proving an exponential inequality by induction** • **Cross-Lesson: 1.04 Partial Fractions**

Prove by induction that $2^n > n$ for all integers $n \geq 1$.

Base case $n = 1$: $2^1 = 2 > 1$, true. Inductive step: assume $2^m > m$ for some $m \geq 1$. Then $2^{m+1} = 2 \cdot 2^m > 2m$ (using the assumption). Since $m \geq 1$, $2m \geq m + 1$, so $2^{m+1} > 2m \geq m + 1$. Thus $2^{m+1} > m + 1$. By induction, $2^n > n$ for all $n \geq 1$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Principle & Architecture of Induction

In a proof by mathematical induction that a proposition $P(n)$ is true for all integers $n \geq 1$, what are the essential logical components?

a. **Base case: prove $P(1)$ is true; Inductive step: prove $P(k) \Rightarrow P(k+1)$; Conclusion by PMI** ✓

b. Verify $P(n)$ for $n = 1, 2, 3, 4, 5$ and deduce the pattern

c. Assume $P(n)$ is true for all n and differentiate both sides

d. Prove $P(k+1) \Rightarrow P(k)$ starting from an arbitrary large integer

The Principle of Mathematical Induction requires establishing the base step (verify $P(1)$), stating the induction hypothesis (assume $P(k)$ is true), showing that $P(k) \Rightarrow P(k+1)$, and stating the formal conclusion.

22 [Foundational] • Target: The Principle & Architecture of Induction

If the base case of an induction proof is verified at $n = 4$, for which set of integers is $P(n)$ established (assuming the inductive step holds for all $k \geq 4$)?

a. $n \in \{4, 5, 6, \ldots\}$ ✓

b. All integers $n \in \mathbb{Z}$

c. $n \in \{1, 2, 3, 4\}$

d. $n \in \mathbb{R}$ with $n \geq 4$

The chain of implications starts at the base case $n = 4$ and propagates to all subsequent integers $n \geq 4$.

23 [Foundational] • Target: The Principle & Architecture of Induction

In proving $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$, what is the algebraic expression for the sum up to $k+1$ using the induction hypothesis?

a. $\frac{k(k+1)}{2} + (k+1)$ ✓

b. $\frac{(k+1)(k+2)}{2} + k$

c. $\frac{k(k+1)}{2} \times (k+1)$

d. $\frac{k(k-1)}{2} + (k+1)$

By definition, $\sum_{r=1}^{k+1} r = \left(\sum_{r=1}^{k} r\right) + (k+1)$. Substituting the inductive hypothesis gives $\frac{k(k+1)}{2} + (k+1)$.

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24 [Foundational] • Target: Summation and Series Proofs

When proving by induction that $\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$, factoring $(k+1)$ from $\frac{k(k+1)(2k+1)}{6} + (k+1)^2$ yields:

a. $\frac{k+1}{6}[2k^2+7k+6] = \frac{(k+1)(k+2)(2k+3)}{6}$ ✓

b. $\frac{k+1}{6}[2k^2+5k+6]$

c. $\frac{(k+1)^2(2k+1)}{6}$

d. $\frac{k+1}{6}[k(2k+1)+1]$

Factor $\frac{k+1}{6}$: $\frac{k+1}{6}[k(2k+1)+6(k+1)] = \frac{k+1}{6}[2k^2+7k+6] = \frac{(k+1)(k+2)(2k+3)}{6}$, which is the target formula at $n = k+1$.

25 [Foundational] • Target: Summation and Series Proofs

For the proposition $\sum_{r=1}^{n} (2r-1) = n^2$, verify the inductive step by simplifying $k^2 + (2(k+1)-1)$.

a. $(k+1)^2$ ✓

b. k^2+2k

c. k^2+2k+3

d. $(k+2)^2$

$k^2 + (2k+2-1) = k^2+2k+1 = (k+1)^2$, completing the induction step.

26 [Foundational] • Target: Summation and Series Proofs

Consider the geometric series $\sum_{r=1}^{n} 2^{r-1} = 2^n - 1$. What is the base case check for $n = 1$?

a. LHS $= 2^0 = 1$, RHS $= 2^1 - 1 = 1$; **LHS = RHS** ✓

b. LHS $= 2^1 = 2$, RHS $= 2^1 - 1 = 1$

c. LHS $= 0$, RHS $= 0$

d. LHS $= 1$, RHS $= 2$

At $n = 1$, LHS $= 2^{1-1} = 1$ and RHS $= 2^1 - 1 = 1$, so the base case is verified.

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27 [Foundational] • Target: Divisibility and Inequality Proofs

To prove $5^n - 1$ is divisible by 4 for $n \geq 1$, express $5^{k+1} - 1$ in terms of $(5^k - 1)$.

a. $5(5^k - 1) + 4$ ✓

b. $5(5^k - 1) - 4$

c. $4(5^k - 1) + 5$

d. $(5^k - 1) + 5$

$5^{k+1} - 1 = 5 \cdot 5^k - 1 = 5(5^k - 1) + 5 - 1 = 5(5^k - 1) + 4$. If $5^k - 1 = 4M$, this equals $5(4M) + 4 = 4(5M + 1)$, which is divisible by 4.

28 [Foundational] • Target: Divisibility and Inequality Proofs

In proving $2^n > n^2$ for $n \geq 5$, assuming $2^k > k^2$, which inequality must be justified to show $2^{k+1} > (k+1)^2$?

a. $2k^2 \geq (k+1)^2 \Leftrightarrow k^2 - 2k - 1 \geq 0$**, which holds for** $k \geq 5$ ✓

b. $k^2 > 2k + 1$ for all $k \in \mathbb{R}$

c. $2^k > 2k$

d. $(k+1)^2 > k^2 + 1$

$2^{k+1} = 2 \cdot 2^k > 2k^2$. We need $2k^2 \geq (k+1)^2 = k^2 + 2k + 1 \Leftrightarrow k^2 - 2k - 1 \geq 0$. For $k \geq 5$, $k(k-2) - 1 \geq 5(3) - 1 = 14 > 0$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Divisibility and Inequality Proofs

To prove $3^{2n} - 1$ is divisible by 8, evaluate the base case $n = 1$.

a. $3^2 - 1 = 8$**, which is divisible by** 8 ✓

b. $3^1 - 1 = 2$

c. $3^0 - 1 = 0$

d. $3^2 - 1 = 9$

At $n = 1$: $3^{2(1)} - 1 = 9 - 1 = 8$, which is divisible by 8 with quotient 1.

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30 [Intermediate] • Target: Calculus and Matrix Applications

Prove by induction that $\frac{d^n}{dx^n}(xe^x) = (x+n)e^x$. What is the inductive step derivative $\frac{d}{dx}[(x+k)e^x]$?

a. $1 \cdot e^x + (x+k)e^x = (x+k+1)e^x$ ✓

b. $(x+k)e^x + e^x = (x+k)e^x$

c. $(x+1)e^x + ke^x$

d. $(x+k+1)xe^x$

By the product rule: $\frac{d}{dx}[(x+k)e^x] = \frac{d}{dx}(x+k)\cdot e^x + (x+k)\cdot e^x = 1\cdot e^x + (x+k)e^x = (x+k+1)e^x$.

31 [Intermediate] • Target: Calculus and Matrix Applications

If $A = [[1,1],[0,1]]$, prove by induction that $A^n = [[1,n],[0,1]]$. What is the matrix product $A^k \cdot A$?

a. $[[1,k],[0,1]][[1,1],[0,1]] = [[1,k+1],[0,1]]$ ✓

b. $[[1,k],[0,1]][[1,1],[0,1]] = [[1,2k],[0,1]]$

c. $[[1,k],[0,1]][[1,1],[0,1]] = [[1,k^2],[0,1]]$

d. $[[1,1],[0,1]]$

Multiplying row by column: $(1)(1)+(k)(0)=1$; $(1)(1)+(k)(1)=k+1$; $(0)(1)+(1)(0)=0$; $(0)(1)+(1)(1)=1$, giving $[[1,k+1],[0,1]]$.

32 [Intermediate] • Target: Calculus and Matrix Applications

A sequence is defined by $u_1 = 5$ and $u_{n+1} = 2u_n - 3$. Prove by induction that $u_n = 2^{n+1} + 3$. Assuming $u_k = 2^{k+1} + 3$, what is u_{k+1}?

a. $2(2^{k+1}+3) - 3 = 2^{k+2} + 6 - 3 = 2^{k+2} + 3$ ✓

b. $2(2^{k+1}) - 3 = 2^{k+2} - 3$

c. $2^{k+1} + 6$

d. $2^{k+2} + 6$

Substitute hypothesis into recurrence: $u_{k+1} = 2u_k - 3 = 2(2^{k+1}+3) - 3 = 2^{k+2} + 6 - 3 = 2^{(k+1)+1} + 3$, completing the inductive step.

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33 [Intermediate] • Target: Induction to prove a formula for a weighted geometric sum

In proving by induction that $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$, assuming the result is true for $n = k$, what must be added to $\frac{k(k+1)}{2}$ to obtain the sum for $n = k+1$?

a. k

b. $k+2$

c. $k+1$ ✓

d. $2k+1$

The $(k+1)$th term of the sum is $k+1$, so we add $k+1$ to reach $\frac{(k+1)(k+2)}{2}$.

34 [Intermediate] • Target: A closing check: identifying an invalid induction argument

Using the first three terms of the Maclaurin series for e^{2x}, approximate $e^{2\times\frac{1}{5}}$ as a fraction.

a. $\frac{59}{50}$

b. $\frac{69}{50}$

c. $\frac{42}{25}$

d. $\frac{37}{25}$ ✓

$e^{2x} \approx 1 + 2x + \frac{(2x)^2}{2}$, at $x = \frac{1}{5}$: $1 + \frac{2}{5} + \frac{\frac{4}{25}}{2} = \frac{37}{25}$

35 [Intermediate] • Target: Proving a sum-of-products formula by induction

Using the first three terms of the Maclaurin series for e^{2x}, approximate $e^{2\times\frac{1}{4}}$ as a fraction.

a. $\frac{13}{8}$ ✓

b. $\frac{57}{40}$

c. $\frac{53}{40}$

d. $\frac{77}{40}$

$e^{2x} \approx 1 + 2x + \frac{(2x)^2}{2}$, at $x = \frac{1}{4}$: $1 + \frac{1}{2} + \frac{\frac{1}{4}}{2} = \frac{13}{8}$

36 [Intermediate] • Target: Proving an exponential inequality by induction

Using the first three terms of the Maclaurin series for e^{x}, approximate $e^{1\times\frac{1}{4}}$ as a fraction.

a. $\frac{173}{160}$

b. $\frac{41}{32}$ ✓

c. $\frac{253}{160}$

d. $\frac{237}{160}$

$e^{x} \approx 1 + x + \frac{(x)^2}{2}$, at $x = \frac{1}{4}$: $1 + \frac{1}{4} + \frac{\frac{1}{16}}{2} = \frac{41}{32}$

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37 [Intermediate] • Target: Proving a divisibility result by induction

Using the first three terms of the Maclaurin series for e^x, approximate $e^{1 \times \frac{1}{5}}$ as a fraction.

a. $\frac{51}{50}$

b. $\frac{38}{25}$

c. $\frac{71}{50}$

d. $\frac{61}{50}$ ✓

$e^x \approx 1 + x + \frac{(x)^2}{2}$, at $x = \frac{1}{5}$: $1 + \frac{1}{5} + \frac{\frac{1}{25}}{2} = \frac{61}{50}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Proving a closed form for a recursive sequence • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Using the first three terms of the Maclaurin series for e^x, approximate $e^{1 \times \frac{1}{2}}$ as a fraction.

a. $\frac{73}{40}$

b. $\frac{13}{8}$ ✓

c. $\frac{57}{40}$

d. $\frac{61}{40}$

$e^x \approx 1 + x + \frac{(x)^2}{2}$, at $x = \frac{1}{2}$: $1 + \frac{1}{2} + \frac{\frac{1}{4}}{2} = \frac{13}{8}$

39 [Challenging] • Target: A second exponential inequality by induction • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

For $x^3 - 3x^2 - 36x + 108 = 0$, find the product of the three roots.

a. -73

b. -140

c. -134

d. **-108** ✓

Working through this complex-numbers step: Product of roots $= -\frac{d}{a} = -(108) = -108$

40 [Challenging] • Target: A third inequality by induction • Cross-Lesson: 1.04 Partial Fractions

For $x^3 - 7x^2 - 14x + 120 = 0$, find the product of the three roots.

a. -128

b. -90

c. -156

d. **-120** ✓

Working through this complex-numbers step: Product of roots $= -\frac{d}{a} = -(120) = -120$

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41 [Challenging] • Target: Induction in combinatorics: counting subsets • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

For $x^3 - 19x - 30 = 0$, find the product of the three roots.

a. **30** ✓

b. 43

c. 27

d. 20

Working through this complex-numbers step: Product of roots $= -\frac{d}{a} = -(-30) = 30$

42 [Challenging] • Target: Proving the sum of the first n odd numbers • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Using the first three terms of the Maclaurin series for e^{2x}, approximate $e^{2 \times \frac{1}{2}}$ as a fraction.

a. $\frac{11}{5}$

b. $\frac{5}{2}$ ✓

c. $\frac{12}{5}$

d. $\frac{14}{5}$

$e^{2x} \approx 1 + 2x + \frac{(2x)^2}{2}$, at $x = \frac{1}{2}$: $1 + 1 + \frac{1}{2} = \frac{5}{2}$

43 [Challenging] • Target: The Principle & Architecture of Induction • Cross-Lesson: 3.01 3D Vectors & the Cross Product

For $x^3 - 7x^2 + 10x = 0$, find the product of the three roots.

a. -1

b. -4

c. **0** ✓

d. -3

Working through this complex-numbers step: Product of roots $= -\frac{d}{a} = -(0) = 0$

44 [Challenging] • Target: Summation and Series Proofs • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. Determine the sum of the x-intercepts.

a. **4** ✓

b. 3

c. 1

d. -3

The x-intercepts of a function are the values of x for which $f(x) = 0$. For the quadratic function $f(x) = x^2 - 4x + 3$, we can find the x-intercepts by solving the equation $x^2 - 4x + 3 = 0$. This is a quadratic equation that can be factored. We are looking for two numbers that multiply to 3 and add to -4. These numbers are -1 and -3. Therefore, the equation can be factored as $(x - 1)(x - 3) = 0$. The solutions are $x = 1$ and $x = 3$. These are the x-intercepts. The sum of the x-intercepts is $1 + 3 = 4$. Alternative.

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45 [Challenging] • Target: Divisibility and Inequality Proofs • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Product of roots of $2x^2 - 5x + 3 = 0$:

a. $\frac{3}{2}$ ✓

b. $\frac{5}{2}$

c. $-\frac{3}{2}$

d. 3

Applying Algebra, Proof & Complex Numbers principles, Product = c/a = 3/2. Therefore, the verified correct option is $\frac{3}{2}$.

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